Student Performance Analytics Dashboard

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Symbiosis Institute of Technology, Pune
Dept. of Applied Science | First Year B.Tech | 2025-26

Introduction

This Python project analyzes student academic data using the Pandas library. The dataset consists of 50 student records with information on Department, Attendance, GPA, Living arrangement, and Gender. The project processes, cleans, and analyzes this data to extract meaningful insights.

Data Manipulation

Pandas & NumPy for efficient data manipulation, analysis, and descriptive statistics.

Analysis Techniques

Categorical analysis, cross-tabulation, grouping, filtering, sorting, and binning.

Preprocessing

Missing value handling, binning, normalization, and encoding for clean analysis.

Objective of the Project

01 Exploratory Data Analysis — Inspect structure, dimensions, data types, and basic statistics.

02 Categorical Data Analysis — Analyze Department, Gender, and Living arrangement distributions.

03 Missing Value Treatment — Handle GPA missing values via mean/median imputation.

04 Data Binning & Formatting — Categorize attendance into Low, Medium, High bins.

05 Normalization & Encoding — Min-Max, Z-score, Label & One-Hot Encoding.

06 Interactive Dashboard — Build multi-panel Plotly analytics dashboard with dropdown controls.

Raw Data

The dataset consists of 50 student records with 6 columns: Student_ID, Department, Attendance, GPA, Living, and Gender. The dataset covers 6 engineering departments. Average GPA: 8.01 | Average Attendance: 79.94%

Department	No. of Students
AIML	9
Robotics	9
Mechanical	8
ENTC	8
Civil	8
CS	8

Summary Statistics

Statistic	Attendance	GPA
Count	50	48
Mean	79.94	8.01
Min	60	6.30
Median	80.50	7.95
Max	97	10.00

Raw Data — Dataset Structure

Each row represents a unique student record. The dataset (raw_data_students.xlsx) has 50 rows × 6 columns with no duplicate records.

Student_ID	Department	GPA	Attendance	Living	Gender
Unique identifier assigned to each student	Academic dept: AIML, Robotics, Mech, ENTC, Civil, CS	Grade Point Average on a scale of 0 to 10	Class attendance percentage (0 to 100)	Living arrangement: Home or PG (Paying Guest)	Gender of the student: Male / Female

Sample Dataset Rows

Student_ID	Department	GPA	Attendance	Living	Gender
Student01	ENTC	9.8	95	PG	Female
Student02	CS	8.7	88	Home	Male
Student03	AIML	7.2	68	Home	Female
Student04	Robotics	8.4	82	PG	Male
Student05	Mechanical	7.0	70	Home	Female
Student10	Civil	— (missing)	75	PG	Male

Raw Data — Cleaning & Validation

Before analysis, data undergoes a thorough validation and cleaning pipeline to ensure all computations are based on clean, reliable data.

Step 1

Detect Missing Values — df.isnull().sum() identifies 2 missing GPA values in Student10 and Student17. All other 5 columns are fully complete.

Step 2

Range Validation — Filter out rows where GPA < 0 or GPA > 10, and where Attendance < 0 or Attendance > 100 using Boolean indexing.

Step 3

Remove Duplicates — df.duplicated().sum() confirms 0 duplicate records. No rows are dropped at this stage — dataset remains at 50 rows.

Step 4

Impute Missing GPA — Missing GPA values imputed with mean GPA (8.01) using fillna(). Ensures dataset stays complete for downstream analysis.

50	48	2	0
Total Records	Valid GPA Records	Missing GPA (imputed)	Duplicates Removed

Libraries & Functions Used

Pandas

Data loading, cleaning, groupby, filtering, binning, and descriptive statistics.

Matplotlib & Seaborn

Base styling for charts via plt.style.use() and sns.set_palette().

Scikit-learn

LabelEncoder to transform categorical Department into numeric codes for 3D plots.

Plotly Express

High-level interactive charts: scatter, scatter_3d, sunburst with trendlines.

Plotly Graph Objects (go)

Fine-grained control: bar, histogram, box, heatmap, and gauge/indicator charts.

make_subplots

Constructs the 2x3 analytics dashboard with six chart types in a unified layout.

Data Inspection

Data inspection is the first step of any analysis pipeline — understanding the structure, size, and properties before any transformations.

```
df = pd.read_csv('raw_data_students.csv')
df.size     # 300 (50 rows x 6 cols)
df.shape    # (50, 6)
df.info()   # Data types & null counts
df.head()   # First 5 rows
df.isnull().sum()   # GPA: 2 missing
df.describe()   # Summary statistics
df.duplicated().sum()   # 0 duplicates
df.nunique()    # Unique values/column
```

Key Findings

- 50 rows, 6 columns
- 0 duplicate records
- 2 missing values in GPA column
- (Student10 & Student17)
- All other columns complete
- Avg GPA: 8.01
- Avg Attendance: 79.94%

Categorical Data Analysis

The three categorical columns — Department, Gender, and Living — were analyzed for distribution and cross-relationships.

50%
Male Students

50%
Female Students

50%
Living at Home

50%
Living in PG

Gender × Department Cross-tabulation

Department	Male	Female
AIML	5	4
Robotics	4	5
Mechanical	4	4
ENTC	3	5
Civil	4	4
CS	5	3

Missing Values & Data Binning

Missing Value Treatment

```
df.isna().sum()        # GPA: 2
df.dropna()            # Drop rows
df.fillna(df.mean())   # Fill mean

# Impute GPA with mean
df['GPA'].fillna(
  df['GPA'].mean(), inplace=True
)
```

GPA missing values (Student10 & Student17) imputed with mean GPA = 8.01

Attendance Binning

Low	60 – 70 →	15 students
Medium	71 – 85 →	20 students
High	86 – 97 →	15 students

Students with High attendance: mainly ENTC & CS departments

Normalization & Encoding

Normalization rescales data for ML models; encoding converts categorical variables to numerical representations.

Min-Max Normalization

Scales GPA to [0, 1] range.
Formula: (x – min) / (max – min)

Z-score Normalization

Standardizes GPA: mean=0, std=1.
Formula: (x – mean) / std

Decimal Scaling

Scales GPA by dividing by 100,000 for uniform magnitude reduction.

Label Encoding

Converts Department names to integer codes (0–5) using sklearn LabelEncoder.

One-Hot Encoding

Creates binary columns for each department. Used for regression/classification.

Dummy Encoding

Like One-Hot but drops the first column to prevent multicollinearity.

Main Dashboard — 2×3 Subplot Layout

A 2×3 interactive Plotly dashboard built with make_subplots() — six visualizations in a unified 1400×800px figure with dropdown controls.

Chart 1: Gender Distribution

Bar chart — coral/teal colors for Male vs Female count.

Chart 2: Dept. GPA Ranking

Grouped bars per department — Set3 palette, groupby mean.

Chart 3: GPA Histogram

15-bin histogram in soft green — reveals GPA distribution shape.

Chart 4: Attendance vs GPA

Scatter + OLS trendline — color-coded by gender.

Chart 5: GPA by Living

Box plot — Home vs PG interquartile GPA comparison.

Chart 6: Correlation Heatmap

Pearson correlation — RdBu_r colorscale, zmid=0.

Advanced Visualizations

3D Scatter Plot

Built with px.scatter_3d() — X: Attendance, Y: GPA, Z: Dept (encoded). Color = Gender, Size = GPA. Enables identification of multi-variate student clusters invisible in 2D charts. Users can rotate, zoom, and isolate gender groups via legend clicks.

Sunburst Chart

4-level hierarchy: Department → Living → Gender → GPA_Category. Built with px.sunburst() + Viridis colorscale. Interactive drill-down reveals proportional relationships across all categorical dimensions simultaneously.

KPI Gauge Charts

8.01
Overall Avg GPA

79.94%
Avg Attendance

9.6
Top Dept (ENTC) GPA

Data Analysis & Results

Multi-layer analysis from descriptive statistics to advanced visualizations yielded the following key findings:

Department	Avg Attendance (%)	Avg GPA	Remarks
AIML	72.8	7.4	Needs attention
Robotics	83.7	8.4	Good
ENTC	93.5	9.6	Top performer ★
CS	87.2	8.9	Strong
Mechanical	71.8	7.1	Intervention req.
Civil	72.4	7.2	Intervention req.

- ENTC students lead with 93.5% attendance & GPA 9.6 — consistent academic excellence.
- Mechanical & Civil departments need targeted academic interventions.
- Gender and Living arrangement perfectly balanced (50-50) — no sampling bias.
- PG students cluster in higher-performing departments (ENTC, CS).

Conclusion

This project demonstrated Python and Pandas' power in extracting actionable insights from student datasets. From loading CSV files to building interactive dashboards, every stage was handled efficiently.

01 ENTC students have the highest avg attendance (93.5%) and GPA (9.6), indicating consistent academic performance.

02 Mechanical and Civil departments show lower attendance and GPA — targeted academic support is recommended.

03 Gender and Living arrangement are perfectly balanced (50-50), confirming no sampling bias in the dataset.

04 Python's normalization, encoding, and binning tools efficiently prepare this dataset for advanced ML applications.